Dynamics Of Machinery

Force Analysis :- 1) Equilibrium condition

a) Force Condition :- $\Sigma F_x = 0$

$\Sigma F_y = 0$

$\Sigma F_z = 0$

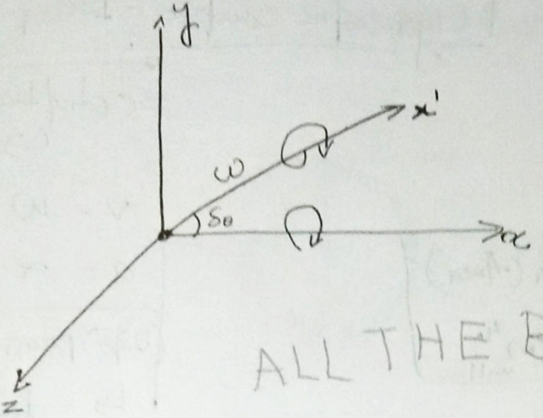
b) Moment condition

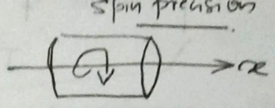
$\hookrightarrow \Sigma M_x = 0$

$\Sigma M_y = 0$

$\Sigma M_z = 0$

Gyroscope



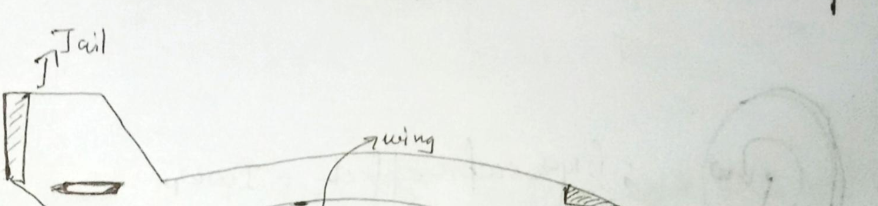


Spin vector direction

CKW - Away

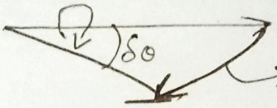
ACKW - Towards

Gyroscopic effect on aircraft



(engine rotation CKW)

→ spin vector ⇒

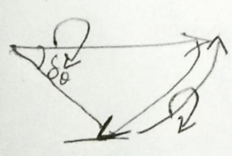


→ Active gyroscopic coupler

↓

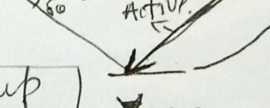
If the spin vector turns right when observed from (top view)

Thus,



Thus if aircraft moves towards right → CKW movement, making the Nose to move down & Tail.

⇒ Tail moves up & nose will dip down.



→ reactive gyroscopic coupler.

↳ when viewed reactive gyroscopic coupler moves away from observer → CKW

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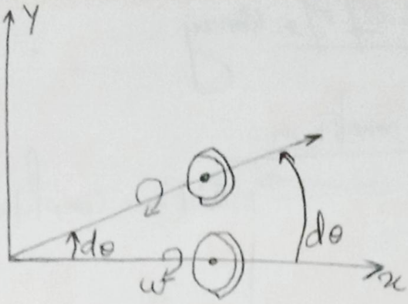

Spin velocity = ω

Change Angular velocity = $\omega \times \frac{d\theta}{dt}$

$\boxed{\text{Angular Acceleration, } \alpha = (\omega \times \omega_p)}$

Torque (couple) = $I \times \alpha$

$\boxed{\text{Gyroscopic couple} = I\omega\omega_p}$

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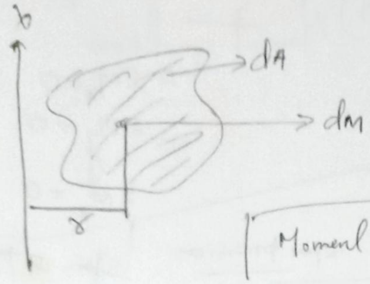


Moment of ~~inter~~ inertia (Area)

$I = r^2 dA$: (m^4) unit.

$I = dm \times r^2$

$I = r^2 dm$ → Mass MOI

s (displacement) ↳ $d\theta$

v = ω

a = α

(ω_p) precision velocity ↳ $= \frac{d\theta}{dt}$

#

; Gyroscopic effect = $I\omega\omega_p$.

$I = mr^2$

$\omega_p = \left(\frac{d\theta}{dt}\right)$

I = mass moment of inertia (mK^2) → here K = radius of gyration

ω = Spin velocity

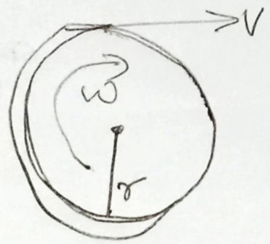
N = rpm

$\omega = \frac{2\pi N}{60}$

ω_p = precision velocity = $\left(\frac{d\theta}{dt}\right)$

Q) An aeroplane flying v = 240 km/hr turns towards left, and completes a quarter circle of $r = 60m$, the mass of rotory engine & propeller of plane is 450 kg with a radius of gyration → 320 mm. The engine speed is 2000 rpm c kw when viewed from rear. Determine the gyroscopic couple on the aircraft & state the effect.

Ans



; $\boxed{V = r\omega}$.

$V = 240\,km/h = 240 \times \frac{5}{18} = 66.66\,m/s$

$r = 60m$

$V = r\omega$

$\omega_p = \frac{V}{r} = \frac{66.66}{60} = 1.111$

$\omega = \frac{2\pi N}{60} = \frac{2 \times \pi \times 2000}{60} = 209.44$ → must be in radian

$K = 320mm$

$m = 450kg$

$I = \cancel{320 \times (450)^2} = 450 \times (320 \times 10^{-3})^2 = \cancel{46.08 \times 10^{13}}\ 46.08$

Gyroscopic couple $= \cancel{4.6}\ I\omega\omega_p$

$= \cancel{4.608 \times 10^{13}}\ 46.08 \times 209.44 \times 1.111$

$= \cancel{1.072 \times 10^{16}}\ 10722.255\,6$ radian

*) Gyroscopic effect on a ship

$\omega_p = \frac{d\theta}{dt} = \phi\omega_0(\cos\omega_0 t)$

$\omega_{p_{max}} = \phi\omega_0 = \omega_p$

$\omega_0 = \frac{2\pi}{T}$

$\alpha = \frac{d}{dt}\omega = -\phi\omega_0^2 \sin\omega t$

$\alpha_{max} = -\phi\omega_0^2$

$\theta = \phi\sin(\omega_0 t)$

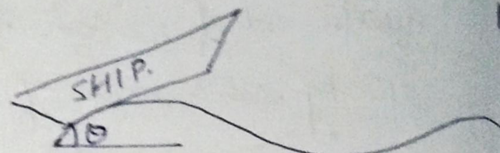
displacement $(x) = X\sin(\omega_0 t)$

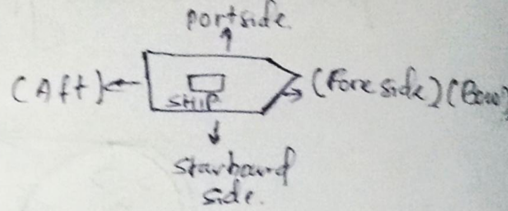
$C_G = I\omega\omega_p$

$= I\omega(\phi\omega_0)$

$C_G = I\omega\phi\left(\frac{2\pi}{T}\right)$

$I = mk^2$





Q) A turbine rotor of a ship has a mass of 2.2 ~~times~~ tonnes and rotates at 1800 rpm ckw when viewed from the Aft (backward). The radius of gyration (k) = 320mm determine the gyroscopic couple & its effect.

i) ship turns right at a radius of 250m with a speed of 25 km/h.

ii) ship pitches with the bow rising with an angular vel of 0.8 rad/sec.

iii) ship rolls at an angular vel of 0.1 rad/sec.

Ans) $m = 2.2$ tonnes $= 2.2\times10^3$ kg.

$N = 1800$ rpm

C.W from Aft

$k = 320$ mm

i) $C_G = ?$

$r = 250$m

$\omega_0 = 25$ km/h ~~→ 25×10³ m/s~~

$\Rightarrow I = mk^2$

$= 2.2\times10^3\times(0.32)^2$

$= 225.28$

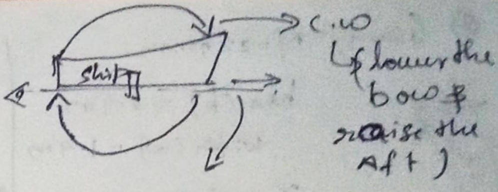
$= 225.3$ kg m^2

$\omega = \frac{2\pi N}{60} = \frac{2\times\pi\times1800}{60} = 188.5$ rad/sec.

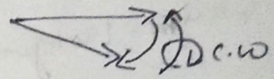
$\omega_p = \frac{V}{R} = \frac{25 \times \frac{5}{18}}{250} = 0.0278$ rad/sec.

$C_G = I\omega\omega_p = 225.3 \times 188.5 \times 0.0278$

$= \underline{1180.64\ Nm}$



when viewed from top view



ii) Pitching of ship

$C_G = I\omega\omega_p \Rightarrow I = 225.28$

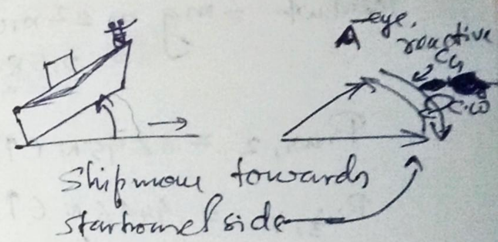
$= 225.3\ kgm^2$

$\omega = 188.5$ rad/sec

given $\omega_p = 0.8$ rad/sec.

$C_G = 225.3 \times 188.5 \times 0.8$

$= 33972\ Nm$; effect is bow raising



iii) rolling

$\boxed{\text{No Gyroscopic effect}}$; only C_G is present

$C_G = I\omega\omega_p$

$= 225.3 \times 188.5 \times 0.1$

$= \underline{4246.9\ Nm}$

Q) A 2.2 tonne racing car has a wheel base of 2.4 m and a track of

(**) DOM (Dynamics of Machinery).

#) Stability of an automobile

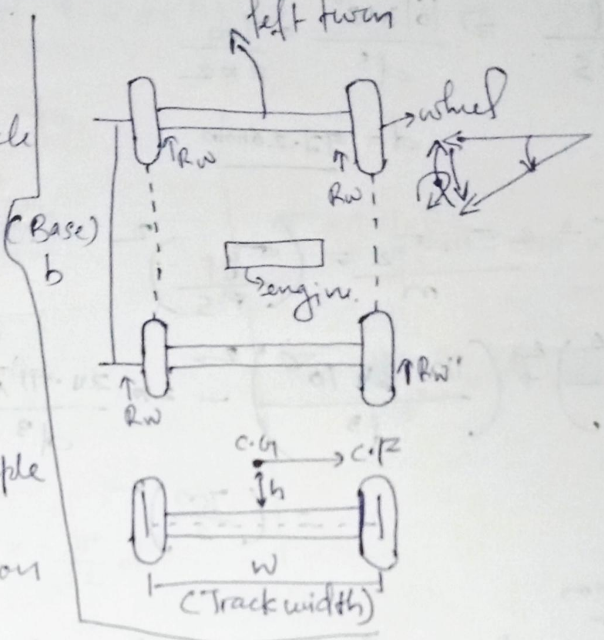
1) Effect due to weight

let w - weight of the vehicle

load on each wheel = $\frac{W}{4}$ (↓)

reaction from the ground

$(R_w) = \frac{W}{4}$ (↑)



2) let C_w - Gyroscopic couple wheels

C_e - Gyroscopic couple on engine

let $G = \frac{C_e}{C_w}$, G = gear ratio $\Rightarrow C_w = I W_w W_p = I \frac{V^2}{rR}$

$W_w = \frac{V}{r}$

$W_p = \frac{V}{R}$

$C_e = I W_e W_p$

$\boxed{C_G = C_w \pm C_e}$

#) effect of C_G on each outer wheel = $\frac{C_G}{2W}$ (downward)

#) effect of C_G on each inner wheel = $\frac{C_G}{2W}$ (upward)

#) reaction of ground on each outer wheel = $\frac{C_G}{2W}$ (upward)

#) " " " " " inner " = $\frac{C_G}{2W}$ (downward)

III) Effect due to centrifugal force

$C.F = mR\omega^2$

$= mR\omega_p^2 = mR\frac{V^2}{R^2}$

$= m\frac{V^2}{R}$

Couple due to C.F = C.F × h

$C_c = \frac{mV^2}{R} \times h.$

#) effect on each inner wheel = $\frac{C_c}{2W}$ (upward)

#) " " " outer " = $\frac{C_c}{2W}$ (downward)

#) Reaction of ground in each inner wheels $[R_c] = \frac{C_c}{2W}$ (downward)

#) " " " " " outer " " = $\frac{C_c}{2W}$ (upward).

#) Total reaction on wheels $= R_W - R_G - R_C$

$$= \frac{W}{4} - \frac{C_G}{2W} - \frac{C_c}{2W}$$

Total reaction on outer wheels $= R_W + R_G + R_C$

$$= \frac{W}{4} + \frac{C_G}{2W} + \frac{C_c}{2W}$$

$$R_W \geq R_G + R_C$$

$$\boxed{\frac{W}{4} \geq \frac{C_G}{2W} + \frac{C_c}{2W}} \longrightarrow \text{Condition of stability.}$$

Q) Given, $M = 22$ tonnes

base $(b) = 2.4m$

Width $(w) = 1.4m$

$r = \frac{0.8}{2} = 0.4$

$R = 100m$

$m = 140 kg$

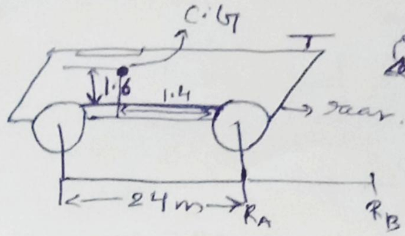
$K = 0.15 m$

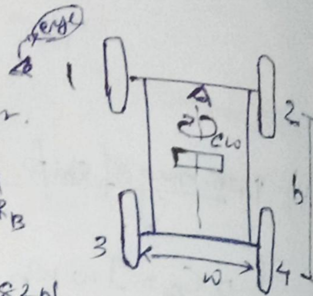
$I_w = 0.7 kgm^2$

$v = 72 km/hr$

$= 72 \times \frac{1000}{3600} = 20 m/s$

given centre of mass lies at 0.6 m above the ground & 1.4m from each rear axis.



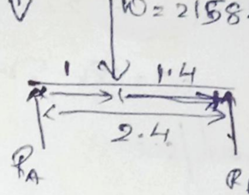


i) reaction due to weight

Total wt $= mg = 22 \times 10^3 \times 9.81$
$= 215820$

$R_{w_{1,2}} = 6295N (\uparrow)$

$R_{w_{3,4}} = 4496 N (\uparrow)$



$\Sigma F_y = 0$

$\boxed{R_A + R_B = 21582N}$

$\Sigma M_z = \Sigma M_A = 0$

$= 21582 \times 1 - 2.4 \times R_B = 0$

$2.4 R_B = 21582$

$R_B = \frac{21582}{2.4}$

$R_B = 8992.5N$

~~$R_A = R_{1\&2} = 1258$~~

$R_A = 21582 - R_B$
$= 12589N$

$R_A = R_{1\&2} = 12589$

$R_2 = R_1 = \frac{1}{2}(R_A)$

$R_{1,2} = \frac{1}{2}(12589) = 6295N$

$R_{3,4} = \frac{1}{2}(R_B)$

$= \frac{1}{2}(89925)$

$= 4496N$

ii) Reaction due to gyroscopic couple.

for wheels

$C_{Gw} = I_w \omega \omega_p$

$= 0.7 \times \left[\frac{V}{r}\right] \times \left[\frac{V}{R}\right]$

$C_{Gw} = 0.7 \times \frac{V^2}{R \times r} = 28 Nm$

$R_{C_{Gw}}$ $(2,4)$ $\downarrow$ $10N (\uparrow)$

$RC_{Gw_{(1,3)}} = \frac{C_{Gw}}{2W} = \frac{28}{2 \times 1.4} = 10N (\uparrow)(\downarrow)$

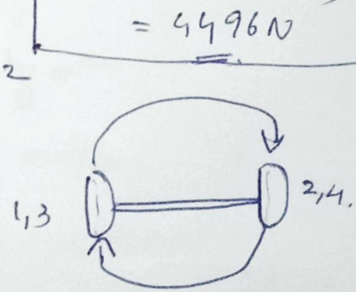
for engine

$C_{Ge} = I_e \omega_e \omega_p$; $I_e = mk^2 = 140 \times 0.15^2$
$= 3.15$

$C_{Ge} = I_e 5 \omega_w \omega_p$

; $\frac{\omega_e}{\omega_w} = 5$

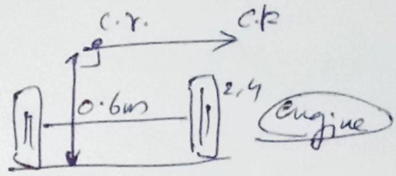
$\omega_e = 5 \times \omega_w$



$\Rightarrow (140 \times 0.15^2) \times 5 \times \frac{v^2}{R \times r} \Rightarrow (140 \times 0.15^2) \times 5 \times \frac{(20)^2}{(0.4 \times 100)} = 157.5\,\text{Nm}$

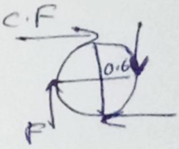
ii) $R_{CGc_{1,2}} = \frac{C_{Ge}}{2b} = \frac{157.5}{2 \times 2.4} = 32.8\ (\uparrow)$

$R_{CGe_{3,4}} = 32.8\,\text{N}\ (\downarrow)$



iii) reaction due to centrifugal couple

$C_c = CF \times 0.6$; Force ⊥ direction = 0.6



$= mR\omega_p^2 \times 0.6$

$= (2.2 \times 10^3) \times 100 \times \frac{20^2}{(100)^2} \times 0.6$

$= 2.2 \times 10 \times 20^2 \times 0.6$

$= 22 \times 40 \times 6$

$= 5280$

$R_{c_{1,3}} = \frac{C_c}{2w}$

$= \frac{5280}{2 \times 1.4} = 1880\,\text{N}\ (\downarrow)$

$R_{c_{2,4}} = 1880\,\text{N}\ (\uparrow)$

$C_c = F \times w$

$F_{1,4} = \frac{C_c}{w}$

Total reaction

wheels 1 = $62915 - 10 + 32.8 - 1880 = 4431.8\,\text{N}$

2 = $62915 + 10 + 32.8 + 1880 = 8223.8\,\text{N}$

3 = $4496 - 10 - 32.8 - 1880 = 2567.2\,\text{N}$

4 = $4496 + 10 - 32.8 + 1880 = 6359.2\,\text{N}$

The four wheels are stable under the given condition.